

INTRO

Hoya Botto

Automation for all Kesennuma manufacturing.

One flap is the first test. A local cooperative is the ambition.

Ron Turetzky · Hackatsuon 2026



TEAM

A team for a local manufacturing automation program



Ron Turetzky

Community builder
Bread Cooperative co-founder
Second Kesennuma Hackatsuon



Decentral Park

Selected organizations in Bread's published portfolio



Stellar



EigenCloud



Funding the Commons



Bread Cooperative

AI and software integration
Interface design and workshops
Manufacturing program lead



FIELDWORK

Hachiyo: our first manufacturing case

A food-packaging visit reveals work between machines and people



株式会社 **ハチヨウ**



Automatic filling



People inspect and close cartons

On 2 October, we visited Hachiyo to understand the packaging work.
Its public account shows automation handing work back to people for checking and closing.
First test: one flap. Then assess handling, inspection and other work across manufacturing.





03 FOLD AND CLOSE

Close the top carton's flaps

3D illustration from fieldwork. Timing and layout simplified.

Three-carton sequence: team fieldwork, 2 Oct 2026; private recording retained. Original 3D illustration; timing simplified, no robot result.

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Public process context

KESENNUMA

The whole manufacturing sector

151 establishments across food, materials, machinery and more

151

manufacturing establishments

78 food + 73 other manufacturing

3,816

people employed

Across all manufacturing categories

¥11.84bn

annual personnel cost

Published 2024 cost base

株式会社 **ハチヨウ**

Food processing

Automatic filling and packing;
people inspect and close cartons.

 **シマ精工**

Precision manufacturing

Machining and production equipment;
assess loading and inspection.

 オフセット印刷・ラベル印刷
双葉印刷株式会社

Printing and finishing

Printing, folding and binding machines;
assess the handling between steps.

Assess what remains manual around existing equipment, sector by sector.

e-Stat: sites / people, 1 Jun 2025; personnel + dispatched-worker cost, 2024. Sole proprietors excluded; company examples are not an adoption survey.

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e-Stat · municipal table



Hachiyo · process



Shima Seiko · equipment

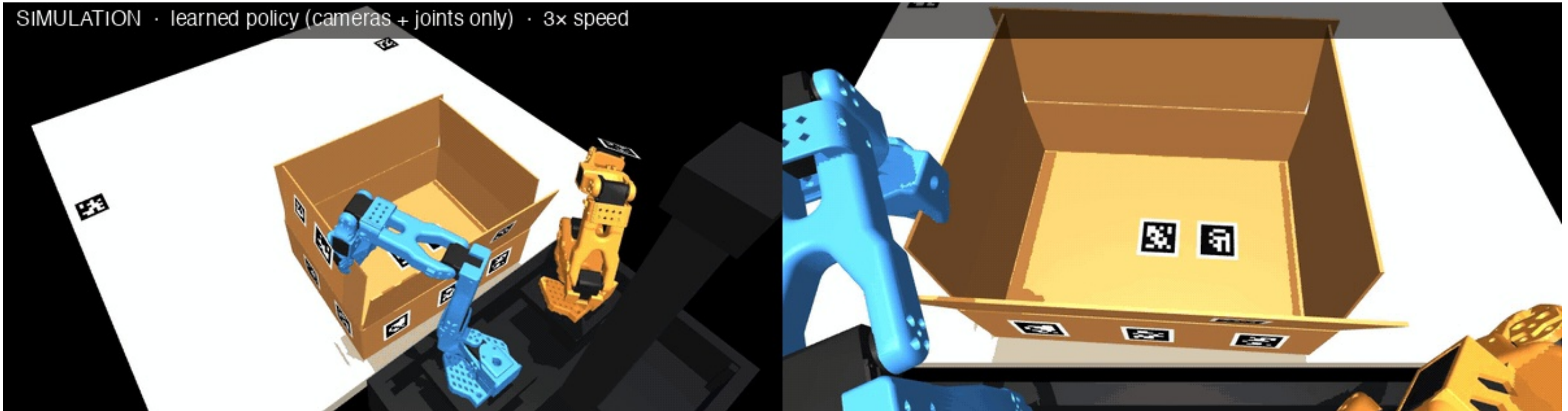


Futaba · factory equipment

DEMO

First test: one box flap

A small test within a manufacturing-wide program

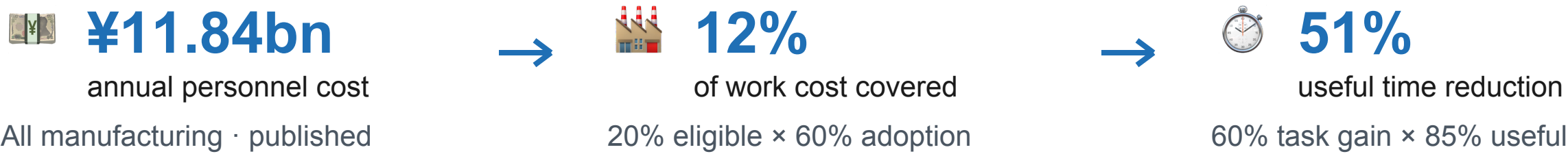


Simulation: a learned policy folds both short flaps using only its cameras and joint angles. It folded 63 of 63 starts it had never seen.
Not yet run on the robot.

ECONOMICS

Potential annual value

Published labor-cost base × work reached × useful time released



REGIONAL SCENARIOS · ANNUAL TIME VALUE BEFORE CAPITAL AND RUNNING COSTS

¥725m

¥11.84bn × 12% × 51%

Broader uptake + task coverage

¥1.79bn

30% × 80% × 70% × 90%

LOCAL APPLICATION FORECASTS · ANNUAL LABOR VALUE, NOT MEASURED SAVINGS

A · Office automation

¥3.67m / year

¥7.68m investment · 76% less process time

C · Washing + capping

¥2.75m / year

¥4.91m investment · 75–88% less process time

D · Seaweed washing

¥5.4m / year

¥11.5m investment · 3 workers redeployed

Regional percentages: our assumptions. Local cases: supplied anonymized application forecasts, 9 Oct 2026; no annual actuals or maintenance costs.



BEYOND BOXES

Beyond boxes, across manufacturing

Every published manufacturing category is in scope; task fit is assessed locally

 Food 78 Food production Inspect · handle · package	 Beverages + feed 10 Beverages, tobacco, feed Fill · label · palletize	 Textiles 7 Clothing + knitwear Inspect · cut · handle	 Materials + print 24 Wood, paper, plastics + more Feed · finish · check quality
 Metals 7 Metal products + nonferrous Load · finish · inspect	 Machinery + electrical 15 Machines + electronics Assemble · test · tend machines	 Transport equipment 8 Shipbuilding + related Fabricate · handle · inspect	 Other manufacturing 2 Other products Choose tasks with each factory

151 establishments

Different processes. Reusable task families.

Feeding · handling · inspection · assembly · finishing · packaging

e-Stat 2025 survey, municipal reference table, rows 1498–1518. 21 reported manufacturing categories grouped into 8; task suggestions are ours.

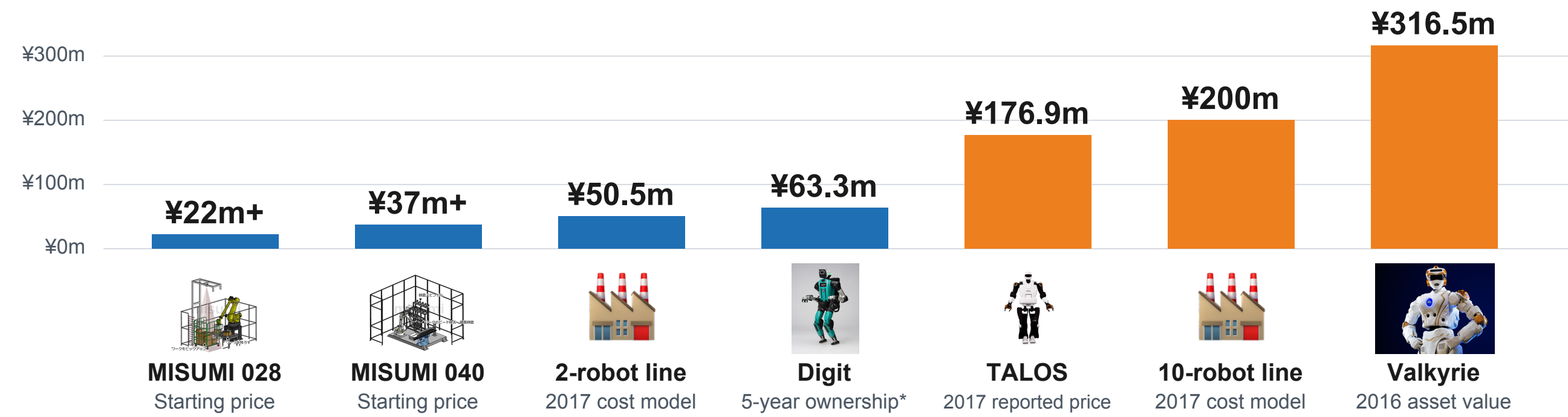


Industrial automation needs serious capital

Local project costs alongside larger system and humanoid references; all amounts in yen

LOCAL APPLICATIONS · 4 ANONYMOUS SEAFOOD CASES
¥4.9m–¥22m per project

Software, processing equipment, washing + capping
Reported project costs; scope differs from the systems below.



Local costs: anonymized applications, 9 Oct 2026. Other systems differ in scope/date. *Digit: five-year ownership model. Historical amounts are not quotes.

 MISUMI 028 · starting

 MISUMI 040 · starting

 JARA 2017 · pp.5–6

 Digit · model pp.40–41

 TALOS · 2017 report

 Valkyrie · 2016 value

 ECB · 7 Oct 2026

 Local applications · forecasts

SHARED DEVELOPMENT

Share automation across manufacturing

Co-ops already work here. Share designs, engineering and support across different tasks.



気仙沼鹿折加工協同組合

Co-ops already work here.

- Shared facilities + people.
- Share engineering + support.
- Measure value kept locally.

151 manufacturing sites · 6 reusable task families



ONE CELL / SITE · PLANNING MODEL



Separate: 151 × ¥50m. Shared: 151 × ¥40m + 6 reusable cores × ¥50m.

Cost comparison: our unquoted model. Local supplier bases do not establish locally retained value. No verified shared-machine savings in the supplied cases.

CITY ALIGNMENT

Built around Kesennuma's priorities

Manufacturing productivity, local enterprise and relationships that continue



生産性向上

Higher productivity

Improve work across manufacturing



新産業・新事業

Local businesses + skills

Train, build and test new services



関係人口

Lasting connections

Return, teach and work together

MEASURES REPORTED IN THE LOCAL APPLICATION SUMMARY

Work time per day

Across the production line, including the process

Gross profit per worker

Part-time hours converted to full-time equivalents

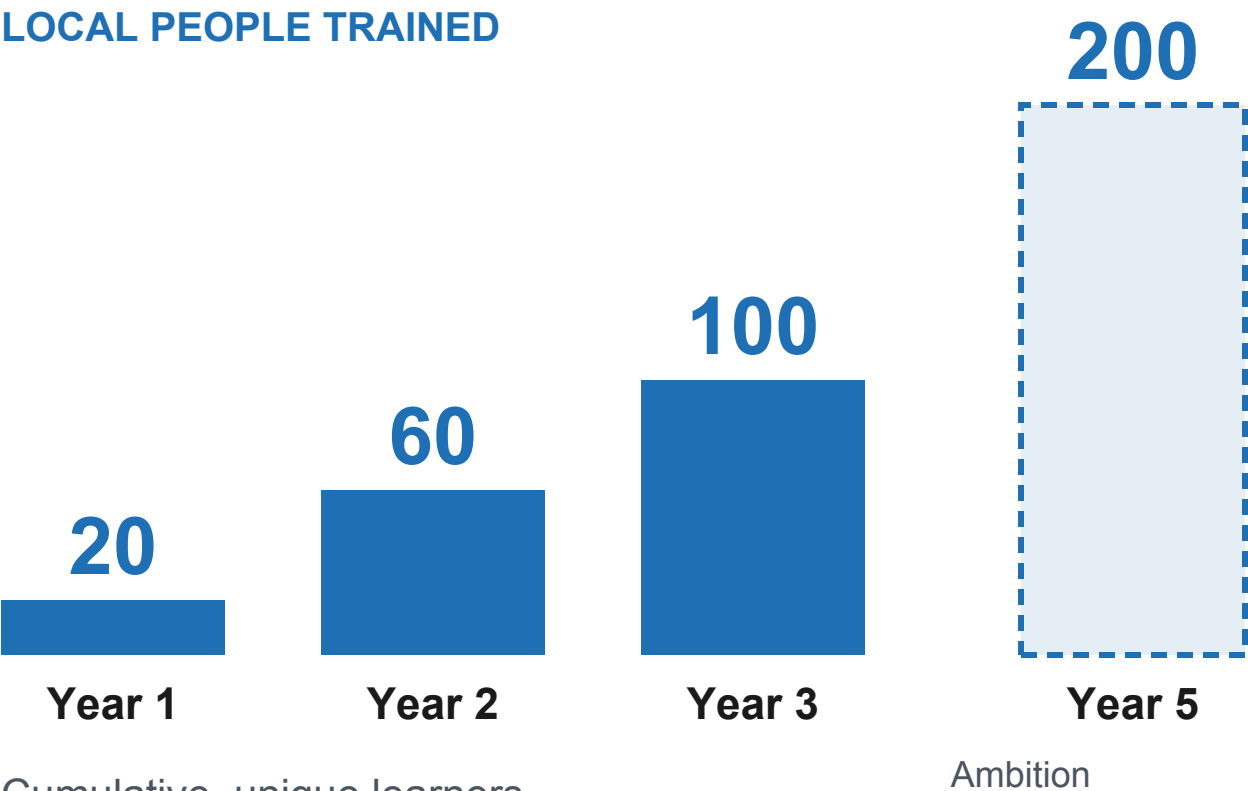


CITY OUTCOMES

Manufacturing skills and local businesses

Operators, technicians and new ventures across the sector

LOCAL PEOPLE TRAINED



Cumulative, unique learners

- 10 Local trainers by year 3**
Graduates teach future cohorts.
Operators, technicians and young makers.
- 20 Factories across manufacturing**
Food, materials, machinery and more.
40 task trials; production needs a quote.
- 5 Local venture pilots by year 3**
Test products and paid services with customers.
Potential new businesses and local inventions.

Years 4–5: further cohorts, local ownership and new businesses if customer demand supports them.

All training, trial and venture numbers: our proposed targets, not published outcomes. Sources below support policy context / team capability only.



WHY NOW

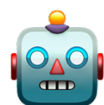
Global robotics is accelerating

AI makes more local manufacturing capability possible. Use the global momentum.

GLOBAL INDUSTRY IS MOVING

+11%

Industrial robot installations
worldwide in 2025
600,000+ new robots



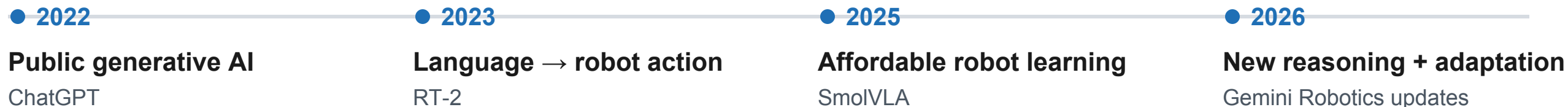
Now possible

Open robot AI can run
on consumer hardware.
More is within small teams' reach.



Use the momentum

Build and learn locally.
Share progress across factories.
Grow skills and new businesses.



Global momentum makes local capability possible.

Primary sources below. Global installations are not local AI adoption. Selected milestones; factory reliability still needs testing.



IFR · 24 Sep 2026



OpenAI · 30 Nov 2022



DeepMind · 28 Jul 2023



SmoIVLA · 3 Jun 2025



ER 1.6 · 14 Apr 2026



Robotics 2 · 30 Jul 2026

ROADMAP

Fund a manufacturing-wide program

A ¥11m pilot, then a ¥300m development and training program if it succeeds

90 DAYS



¥11m

Pilot + handover

One flap; measure the value
Train 1 lead + 2 operators

YEAR 1



¥60m

Try other processes

20 learners · 3 sites
6 trials across task families

YEAR 2



¥100m

Reuse across sectors

60 learners · 10 sites
20 cumulative task trials

YEAR 3



¥140m

Local services + ventures

100 learners · 20 sites
40 trials · 5 venture pilots

¥300m follow-on program

Development, trials, training and venture support

¥100m/year savings target

40 production tasks × ¥2.5m net annual saving

Production capex is additional. Local case forecasts do not establish net cash savings.

Year 5 ambition: 200 learners and locally run services; further funding required.

Proposed budgets and conditional targets. Production capex is additional. Case forecasts are gross labor value; maintenance and actual results are unknown.

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Bread · team capability



Local applications · forecasts